



Sri Chaitanya IIT Academy.,India.

A.P. T.S. KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S) & LIIT

JEE-MAIN

Date: 24-12-2025

Time: 09.00Am to 12.00Pm

GTM-04

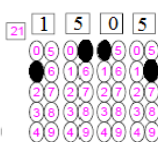
Max. Marks: 300

IMPORTANT INSTRUCTION:

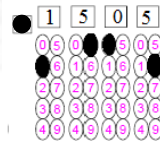
1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black Ball Point Pen** only.
 2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
 3. The test is of **3 hours** duration.
 4. The Test Booklet consists of **75 Questions**. The maximum marks are **300**.
 5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics and Chemistry** having **25 Questions** in each subject and subject having **two sections**.
(I) Section-I contains **20 Multiple Choice Questions** with only one correct option.
Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
(II) Section-II contains **05 Numerical Value Type Questions**.
- The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking



Question Cancelled for Marking

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____

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**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

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RANK 1

341

RANK

1

JEE Advanced 2023

VAVILALA CHIVILAS REDDY

RANK 1

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1

NEET 2023

BORA VARUN CHAKRAVARTI

RANK 1

722

RANK

1

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MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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**MATHEMATICS****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

1. Consider the matrix $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Given below are two statements :

Statement I : $f(-x)$ is the inverse of the matrix $f(x)$

Statement II : $f(x)f(y) = f(x+y)$.

In the light of the above statements, choose the correct answer from the options given below.

- 1) Both Statement I and Statement II are true.
 - 2) Statement I is true but Statement II is false.
 - 3) Both Statement I and Statement II are false.
 - 4) Statement I is false but Statement II is true.
2. The length of the chord of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$, whose mid point is $\left(1, \frac{2}{5}\right)$, is equal to :
- 1) $\frac{\sqrt{1691}}{5}$
 - 2) $\frac{\sqrt{1541}}{5}$
 - 3) $\frac{\sqrt{2009}}{5}$
 - 4) $\frac{\sqrt{1741}}{5}$
3. If $\int_0^1 \frac{1}{\sqrt{3+x} + \sqrt{1+x}} dx = a + b\sqrt{2} + c\sqrt{3}$, where a, b, c are rational numbers, then $2a+3b-4c$ is equal to :
- 1) 8
 - 2) 4
 - 3) 10
 - 4) 7
4. If $a = \lim_{x \rightarrow 0} \frac{\sqrt{1+\sqrt{1+x^4}} - \sqrt{2}}{x^4}$ and $b = \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1+\cos x}}$, then the value of ab^3 is :
- 1) 32
 - 2) 36
 - 3) 30
 - 4) 25

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ALL INDIA**RANK****1**



5. Consider the function $f(x) = \begin{cases} \frac{a(7x-12-x^2)}{b|x^2-7x+12|} & , x < 3 \\ 2 \frac{\sin(x-3)}{x-[x]} & , x > 3 \\ b & , x = 3 \end{cases}$, Where $[x]$ denotes greatest integer less than or equal to x . If S denotes the set of all ordered pairs (a,b) such that $f(x)$ is continuous at $x=3$, then the number of elements in S is :

1) 2 2) 1 3) Infinitely many 4) 4

6. Let $a_1, a_2, \dots, a_{10}$ be 10 observations such that $\sum_{k=1}^{10} a_k = 50$ and $\sum_{\forall k < j} a_k \cdot a_j = 1100$. Then the standard deviation of $a_1, a_2, \dots, a_{10}$ is equal to :

1) $\sqrt{5}$ 2) 10 3) $\sqrt{115}$ 4) 5

7. If the shortest distance of the parabola $y^2 = 4x$ from the centre of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$ is d , then d^2 is equal to :

1) 16 2) 20 3) 36 4) 24

8. The distance, of the point $(7, -2, 11)$ from the line $\frac{x-6}{1} = \frac{y-4}{0} = \frac{z-8}{3}$ along the line $\frac{x-5}{2} = \frac{y-1}{-3} = \frac{z-5}{6}$, is:

1) 21 2) 12 3) 18 4) 14

9. If the shortest distance between the lines $\frac{x-4}{1} = \frac{y+1}{2} = \frac{z}{-3}$ and $\frac{x-\lambda}{2} = \frac{y+1}{4} = \frac{z-2}{-5}$ is $\frac{6}{\sqrt{5}}$, then the sum of all possible values of λ is :

1) 10 2) 7 3) 8 4) 5

10. ${}^{n-1}C_r = (k^2 - 8) {}^nC_{r+1}$ if and only if :

1) $2\sqrt{2} < k \leq 3$ 2) $2\sqrt{2} < k < 2\sqrt{3}$ 3) $2\sqrt{3} < k < 3\sqrt{3}$ 4) $2\sqrt{3} < k \leq 3\sqrt{2}$





11. Let $x=x(t)$ and $y=y(t)$ be solutions of the differential equations $\frac{dx}{dt} + ax = 0$ and $\frac{dy}{dt} + by = 0$ respectively, $a, b \in \mathbb{R}$. Given that $x(0)=2$; $y(0)=1$ and $3y(1)=2x(1)$, the value of t , for which $x(t)=y(t)$, is:
- 1) $\log_2 \frac{2}{3}$ 2) $\log_3 4$ 3) $\log_4 3$ 4) $\log_4 \frac{2}{3}$
12. Let $\vec{a} = \hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = 3(\hat{i} - \hat{j} + \hat{k})$. Let $\vec{c}$ be the vector such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$. Then $\vec{a} \cdot ((\vec{c} \times \vec{b}) - \vec{b} - \vec{c})$ is equal to :
- 1) 20 2) 24 3) 36 4) 32
13. The portion of the line $4x+5y=20$ in the first quadrant is trisected by the lines L_1 and L_2 passing through the origin. The tangent of an angle between the lines L_1 and L_2 is:
- 1) $25/41$ 2) $30/41$ 3) $8/5$ 4) $2/5$
14. If (a, b) be the orthocenter of the triangle whose vertices are $(1, 2)$, $(2, 3)$ and $(3, 1)$ and $I_1 = \int_a^b x \sin(4x - x^2) dx$, $I_2 = \int_a^b \sin(4x - x^2) dx$, then $36 \frac{I_1}{I_2}$ is equal to :
- 1) 72 2) 88 3) 66 4) 80
15. If A denotes the sum of all the coefficients in the expansion of $(1 - 3x + 10x^2)^n$ and B denotes the sum of all the coefficients in the expansion of $(1 + x^2)^n$, then :
- 1) $A = B^3$ 2) $B = A^3$ 3) $3A = B$ 4) $A = 3B$
16. The number of common terms in the progressions $4, 9, 14, 19, \dots$, up to 25^{th} term and $3, 6, 9, 12, \dots$, up to 37^{th} term is :
- 1) 5 2) 8 3) 7 4) 9





17. Let $S = \{1, 2, 3, \dots, 10\}$. Suppose M is the set of all the subsets of S , then the relation $R = \{(A, B) : A \cap B \neq \phi; A, B \in M\}$ is :
- 1) symmetric and transitive only 2) reflexive only
3) symmetric and reflexive only 4) symmetric only
18. Four distinct point $(2k, 3k)$, $(1, 0)$, $(0, 1)$ and $(0, 0)$ lie on a circle for k equal to:
- 1) $3/13$ 2) $1/13$ 3) $2/13$ 4) $5/13$
19. If $S = \{z \in \mathbb{C} : |z - i| = |z + i| = |z - 1|\}$, then, $n(S)$ is :
- 1) 2 2) 3 3) 1 4) 0
20. The function $f: \mathbb{N} - \{1\} \rightarrow \mathbb{N}$; defined by $f(n)$ = the highest prime factor of n , is :
- 1) neither one-one nor onto 2) Onto only
3) both one-one and onto 4) One-one only

SECTION-II (NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Let for a differentiable function $f: (0, \infty) \rightarrow \mathbb{R}$, $f(x) - f(y) \geq \log_e \left(\frac{x}{y} \right) + x - y, \forall x, y \in (0, \infty)$. Then $\sum_{n=1}^{20} f' \left(\frac{1}{n^2} \right)$ is equal to _____
22. If $8 = 3 + \frac{1}{4}(3+p) + \frac{1}{4^2}(3+2p) + \frac{1}{4^3}(3+3p) + \dots \infty$, then the value of p is _____
23. Let the area of the region $\{(x, y) : x - 2y + 4 \geq 0, x + 2y^2 \geq 0, x + 4y^2 \leq 8, y \geq 0\}$ be $\frac{m}{n}$, where m and n are coprime numbers. Then $m+n$ is equal to _____
24. If the solution of the differential equation $(2x + 3y - 2)dx + (4x + 6y - 7)dy = 0, y(0) = 3$, is $\alpha x + \beta y + 3 \log_e |2x + 3y - \gamma| = 6$, then $\alpha + 2\beta + 3\gamma$ is equal to _____
25. A fair die is tossed repeatedly until a six is obtained. Let X denote the number of tosses required and let $a = P(X = 3), b = P(X \geq 3)$ and $c = P(X \geq 6 | X > 3)$. Then $\frac{b+c}{a}$ is equal to _____

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**PHYSICS****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

26. In a linear Simple Harmonic Motion (SHM)
 (A) Restoring force is directly proportional to the displacement
 (B) The acceleration and displacement are opposite in direction
 (C) The velocity is maximum at mean position
 (D) The acceleration is minimum at extreme points.
 Choose the correct answer from the options given below:
 1) (A), (B) and (D) only 2) (A), (B) and (C) only
 3) (C) and (D) only 4) (A), (C) and (D) only
27. The half-life of a radioactive nucleus is 5 years. The fraction of the original sample that would decay in 15 years is:
 1) $1/4$ 2) $1/8$ 3) $7/8$ 4) $3/4$
28. The de Broglie wavelength of an electron having kinetic energy E is λ . If the kinetic energy of electron becomes $E/4$, then its de-Broglie wavelength will be:
 1) $\frac{\lambda}{\sqrt{2}}$ 2) $\frac{\lambda}{2}$ 3) $\sqrt{2}\lambda$ 4) 2λ
29. A single slit of width a is illuminated by a monochromatic light of wavelength 600nm . The value of ' a ' for which first minimum appears at $\theta=30^\circ$ on the screen will be:
 1) $3\mu\text{m}$ 2) $1.2\mu\text{m}$ 3) $1.8\mu\text{m}$ 4) $0.6\mu\text{m}$
30. Match List I with List II of Electromagnetic waves with corresponding wavelength range:

	List I		List II
A)	Microwave	I)	400 nm to 1 nm
B)	Ultraviolet	II)	1 nm to 10^{-3} nm
C)	X-Ray	III)	1 mm to 700 nm
D)	Infra-red	IV)	0.1 m to 1 mm

Choose the correct answer from the options given below:

- 1) A-IV, B-II, C-I, D-III 2) A-IV, B-I, C-III, D-II
 3) A-I, B-IV, C-II, D-III 4) A-IV, B-I, C-II, D-III

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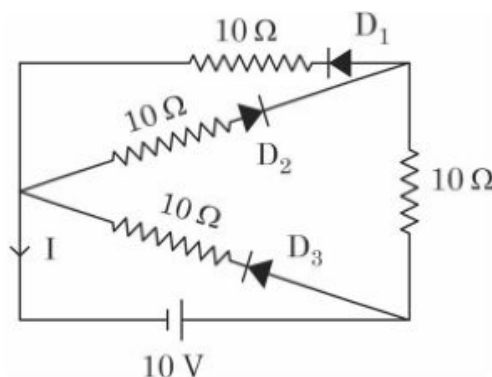


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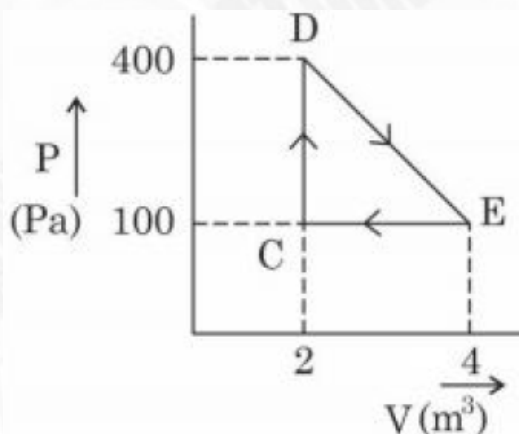


31. In the given circuit, the current (I) through the battery will be



- 1) 1.5 A 2) 2.5 A 3) 2A 4) 1 A

32. A thermodynamic system is taken through cyclic process. The total work done in the process is:



- 1) 100 J 2) 300 J 3) Zero 4) 200 J

33. Given below are two statements:

Statement I: An AC circuit undergoes electrical resonance if it contains either a capacitor or an inductor.

Statement II: An AC circuit containing a pure capacitor or a pure inductor consumes high power due to its non-zero power factor.

In the light of above statements, choose the correct answer from the options given below:

- 1) Both statement I and Statement II are true
 2) Statement I is false but statement II is true
 3) Both statement I and Statement II are false
 4) Statement I is true but statement II is false

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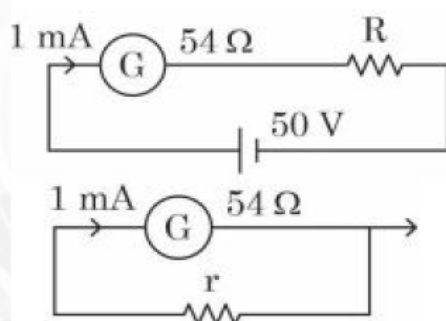
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34. The position vector of a particle related to time t is given by $\vec{r} = (10t\hat{i} + 15t^2\hat{j} + 7t\hat{k})\text{m}$. The direction of net force experienced by the particle is:
- 1) In x-y plane 2) Positive y-axis 3) Positive x-axis 4) Positive z-axis
35. A vector in x - y plane makes an angle of 30° with y-axis. The magnitude of y-component of vector is $2\sqrt{3}$. The magnitude of x-component of the vector will be:
- 1) $\sqrt{3}$ 2) $\frac{1}{\sqrt{3}}$ 3) 6 4) 2
36. For designing a voltmeter of range 50 V and an ammeter of range 10 mA using a galvanometer which has a coil of resistance 54Ω showing a full-scale deflection for 1 mA as in figure.



- (A) For voltmeter $R \approx 50\text{k}\Omega$ (B) For ammeter $r \approx 0.2\Omega$
 (C) For ammeter $r = 6\Omega$ (D) For voltmeter $R \approx 5\text{k}\Omega$
 (E) For voltmeter $R \approx 500\Omega$

Choose the correct answer from the options given below:

- 1) (A) and (B) 2) (C) and (D) 3) (A) and (C) 4) (C) and (E)
37. Given below are two statements:
- Statement I :** The equivalent resistance of resistors in a series combination is smaller than least resistance used in the combination.
- Statement II:** The resistivity of the material is independent of temperature.
- In the light of the above statement, choose the correct answer from the option given below
- 1) Statement I is false but Statement II is true.
 2) Statement I is true but Statement II is false.
 3) Both Statement I and Statement II are true.
 4) Both Statement I and Statement II are false





38. The electric field due to a short electric dipole at a large distance (r) from center of dipole on the equatorial plane varies with distance as:
 1) r 2) $1/r$ 3) $1/r^3$ 4) $1/r^2$
39. A wire of length ' L ' and radius ' r ' is clamped rigidly at one end. When the other end of the wire is pulled by a force f , its length increases by ℓ . Another wire of same material of length ' $2L$ ' and radius ' $2r$ ' is pulled by a force ' $2f$ '. Then the increase in its length will be:
 1) 2ℓ 2) ℓ 3) $\ell/2$ 4) 4ℓ
40. A body is released from a height equal to the radius (R) of the earth. The velocity of the body when it strikes the surface of the earth will be:
 (Given g = acceleration due to gravity on the earth)
 1) $\sqrt{2gR}$ 2) $\sqrt{gR}$ 3) $\sqrt{4gR}$ 4) $\sqrt{\frac{gR}{2}}$
41. The speed of a wave produced in water is given by $v = \lambda^a g^b \rho^c$. Where λ , g and ρ are wavelength of wave, acceleration due to gravity and density of water respectively. The values of a , b and c respectively, are:
 1) $1/2, 0, 1/2$ 2) $1/2, 1/2, 0$ 3) $1, 1, 0$ 4) $1, -1, 0$
42. The position of a particle related to time is given by $x = (5t^2 - 4t + 5)m$. The magnitude of velocity of the particle at $t = 2s$ will be:
 1) 06 ms^{-1} 2) 10 ms^{-1} 3) 16 ms^{-1} 4) 14 ms^{-1}
43. A flask contains Hydrogen and Argon in the ratio 2:1 by mass. The temperature of the mixture is 30°C . The ratio of average kinetic energy per molecule of the two gases ($K_{\text{argon}} / K_{\text{hydrogen}}$) is
 (Given: Atomic Weight of Ar = 39.9)
 1) 2 2) 1 3) $39.9/2$ 4) 39.9
44. Two identical particles each of mass ' m ' go round a circle of radius a under the action of their mutual gravitational attraction. The angular speed of each particle will be:
 1) $\sqrt{\frac{Gm}{a^3}}$ 2) $\sqrt{\frac{Gm}{4a^3}}$ 3) $\sqrt{\frac{Gm}{8a^3}}$ 4) $\sqrt{\frac{Gm}{2a^3}}$
45. A block of mass 10 kg is moving along x -axis under the action of force $F = 5x \text{ N}$. The work done by the force in moving the block from $x = 2\text{m}$ to 4m will be ____ J.
 1) 30 2) 4.2 3) 3.5 4) 6.5





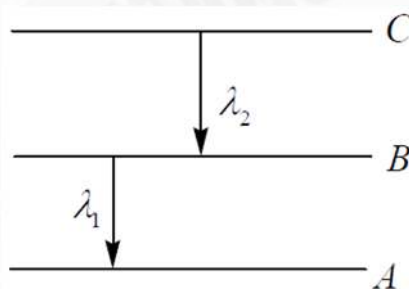
SECTION-II

(NUMERICAL VALUE TYPE)

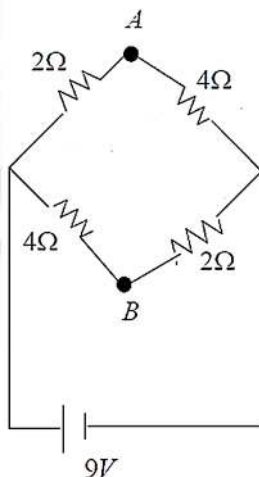
This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. There is an air bubble of radius 1.0 mm in a liquid of surface tension 0.075 Nm^{-1} and density 1000 kg m^{-3} at a depth of 10 cm below the free surface. The amount by which the pressure inside the bubble is greater than the atmospheric pressure is ____ Pa. ($g = 10 \text{ ms}^{-2}$)
47. The fundamental frequency of vibration of a string stretched between two rigid support is 50 Hz. The mass of the string is 18 g and its linear mass density is 20 g/m. The speed of the transverse waves so produced in the string is ____ ms^{-1} .
48. As per given figure A, B and C are the first, second and third excited energy levels of hydrogen atom respectively. If the ratio of the two wavelengths $\left(\text{i.e. } \frac{\lambda_1}{\lambda_2} \right)$ is $\frac{7}{4n}$, then the value of n will be ____.



49. The refractive index of a transparent liquid filled in an equilateral hollow prism is $\sqrt{2}$. The angle of minimum deviation for the liquid will be ____°.
50. A network of four resistance is connected to 9V battery, as shown in figure. Then magnitude of voltage difference between the points A and B is ____ V.



**CHEMISTRY****Max Marks: 100****SECTION-I
(SINGLE CORRECT ANSWER TYPE)**

This section contains **20 Multiple Choice Questions**. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

51. The elemental composition of a compound is 54.2% C, 9.2% H and 36.6% O. If the molar mass of the compound is 132 g mol^{-1} , the molecular formula of the compound is :
[Given : The relative atomic mass of C : H : O = 12 : 1 : 16]
1) $\text{C}_4\text{H}_9\text{O}_3$ 2) $\text{C}_6\text{H}_{12}\text{O}_6$ 3) $\text{C}_6\text{H}_{12}\text{O}_3$ 4) $\text{C}_4\text{H}_8\text{O}_2$
52. When Ethane-1,2-diamine is added progressively to an aqueous solution of Nickel (II) chloride, the sequence of colour change observed will be :
1) Pale Blue $\rightarrow$ Blue $\rightarrow$ Green $\rightarrow$ Violet
2) Pale Blue $\rightarrow$ Blue $\rightarrow$ Violet $\rightarrow$ Green
3) Green $\rightarrow$ Pale Blue $\rightarrow$ Blue $\rightarrow$ Violet
4) Violet $\rightarrow$ Blue $\rightarrow$ Pale Blue $\rightarrow$ Green
53. The conditions and consequence that favours the t_{2g}^3, e_g^1 configuration in a metal complex are :
1) weak field ligand, high spin complex
2) strong field ligand, high spin complex
3) strong field ligand, low spin complex
4) weak field ligand, low spin complex
54. Identify correct statement/s :
(A) $-\text{OCH}_3$ and $-\text{NHCOCH}_3$ are activating group
(B) $-\text{CN}$ and $-\text{OH}$ are meta directing group
(C) $-\text{CN}$ and $-\text{SO}_3\text{H}$ are meta directing group
(D) Activating groups act as ortho and para directing groups
(E) Halides are activating groups
Choose the **correct** answer from the options given below :
1) (A), (C) and (D) only 2) (A), (B) and (E) only
3) (A) only 4) (A) and (C) only

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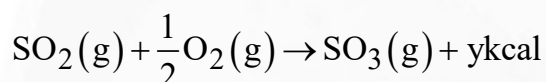
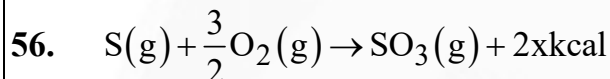
55. Given below are two statements :

Statement (I): The first ionization energy of Pb is greater than that of Sn

Statement(II): The first ionization energy of Ge is greater than that of Si.

In the light of the above statements, choose the correct answer from the options given below:

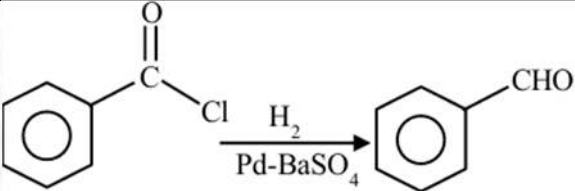
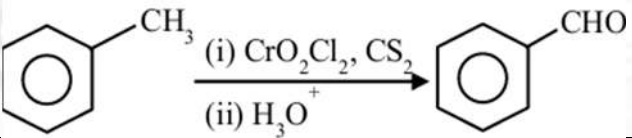
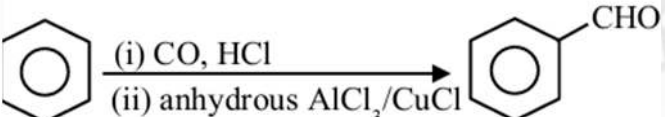
- 1) Statement I is true but Statement II is false.
- 2) Both Statement I and Statement II are false.
- 3) Statement I is false but Statement II is true.
- 4) Both Statement I and Statement II are true.



The heat of formation of $SO_2(g)$ is given by :

- 1) $\frac{2x}{y} \text{ kcal}$
- 2) $y - 2x \text{ kcal}$
- 3) $2x + y \text{ kcal}$
- 4) $x + y \text{ kcal}$

57. Match List-I with List-II.

	List-I		List-II
(A)	$RCN \xrightarrow[(ii) H_3O^+]{(i) SnCl_2, HCl} RCHO$	(I)	Etard reaction
(B)		(II)	Gatterman –Koch reaction
(C)		(III)	Rosenmund reduction
(D)		(IV)	Stephen reaction

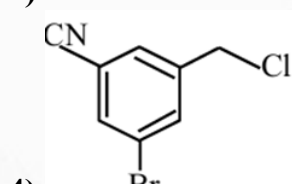
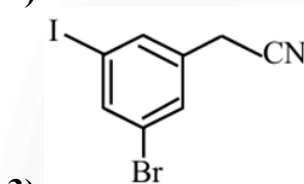
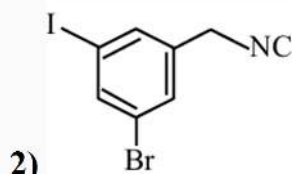
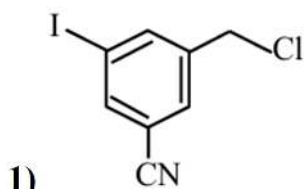
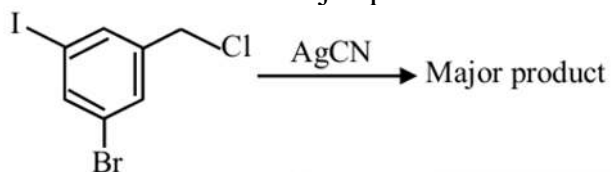
Choose the correct answer from the options given below:

- 1) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
- 2) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
- 3) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
- 4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)





58. The structure of the major product formed in the following reaction is :



59. Match List-I with List-II.

	List-I		List-II
(A)	Adenine	(I)	
(B)	Cytosine	(II)	
(C)	Thymine	(III)	
(D)	Uracil	(IV)	

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Choose the correct answer from the options given below :

1) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

3) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

60. The successive 5 ionisation energies of an element are 800, 2427, 3658, 25024 and 32824 kJ/mol, respectively. By using the above values predict the group in which the above element is present :

1) Group 2

2) Group 13

3) Group 4

4) Group 14

61. Based on the data given below:

$$E^0_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+}} = 1.33\text{V} \quad E^0_{\text{Cl}_2/\text{Cl}^{(-)}} = 1.36\text{V}$$

$$E^0_{\text{MnO}_4^-/\text{Mn}^{2+}} = 1.51\text{V} \quad E^0_{\text{Cr}^{3+}/\text{Cr}} = -0.74\text{V}$$

the strongest reducing agent is :

1) Mn^{2+}

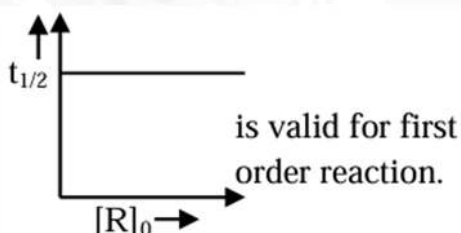
2) Cr

3) MnO_4^-

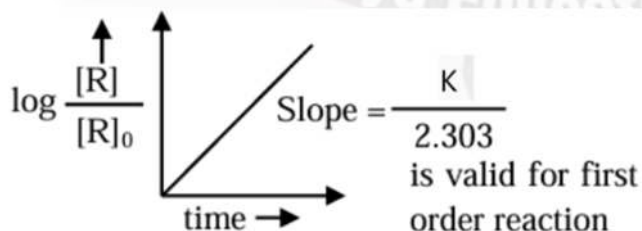
4) Cl^-

62. Given below are two statements :

Statement(I) :



Statement(II) :



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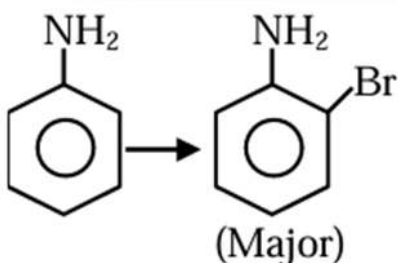
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In light of the above statements, choose the correct answer from the options given below:

- 1) Both Statement I and Statement II are false.
- 2) Statement I is false but Statement II is true.
- 3) Both Statement I and Statement II are true.
- 4) Statement I is true but Statement II is false.

63. For reaction



The correct order of set of reagents for the above conversion is :

- 1) $\text{Br}_2 \mid \text{FeBr}_3, \text{H}_2\text{O}(\Delta), \text{NaOH}$
- 2) $\text{H}_2\text{SO}_4, \text{Ac}_2\text{O}, \text{Br}_2, \text{H}_2\text{O}(\Delta), \text{NaOH}$
- 3) $\text{Ac}_2\text{O}, \text{Br}_2, \text{H}_2\text{O}(\Delta), \text{NaOH}$
- 4) $\text{Ac}_2\text{O}, \text{H}_2\text{SO}_4, \text{Br}_2, \text{NaOH}$

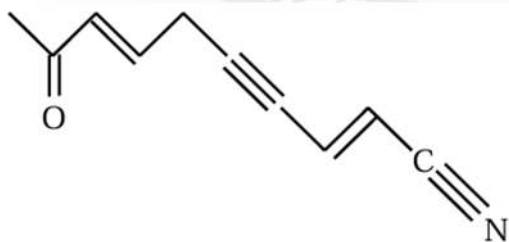
64. For hydrogen atom, the orbital/s with lowest energy is/are :

- (A) 4s (B) 3p_x (C) 3d_{x²-y²} (D) 3d_{z²} (E) 4p_z

Choose the correct answer from the options given below:

- 1) (A) and (E) only
- 2) (B) only
- 3) (A) only
- 4) (B), (C) and (D) only

65. In the given structure, number of sp and sp² hybridized carbon atoms present respectively are:



- 1) 3 and 6
- 2) 3 and 5
- 3) 4 and 6
- 4) 4 and 5

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66. Which of the following mixing of 1M base and 1M acid leads to the largest increase in temperature?

- 1) 30 mL HCl and 30 mL NaOH 2) 30 mL CH₃COOH and 30 mL NaOH
3) 50 mL HCl and 20 mL NaOH 4) 45 mL CH₃COOH and 25 mL NaOH

67. Given below are two statements :

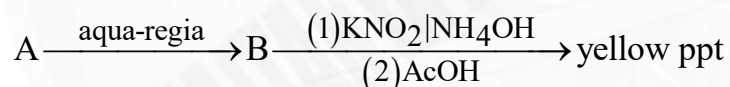
Statement(I) : Experimentally determined oxygen-oxygen bond lengths in the O₃ are found to be same and the bond length is greater than that of a O = O (double bond) but less than that of a single (O – O) bond.

Statement (II) : The strong lone pair-lone pair repulsion between oxygen atoms is solely responsible for the fact that the bond length in ozone is smaller than that of a double bond (O=O) but more than that of a single bond (O – O).

In the light of the above statements, choose the correct answer from the options given below:

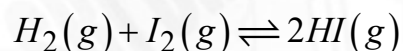
- 1) Statement I is true but Statement II is false.
2) Both Statement I and Statement II are true.
3) Both Statement I and Statement II are false.
4) Statement I is false but Statement II is true.

68. Find the compound 'A' from the following reaction sequences.

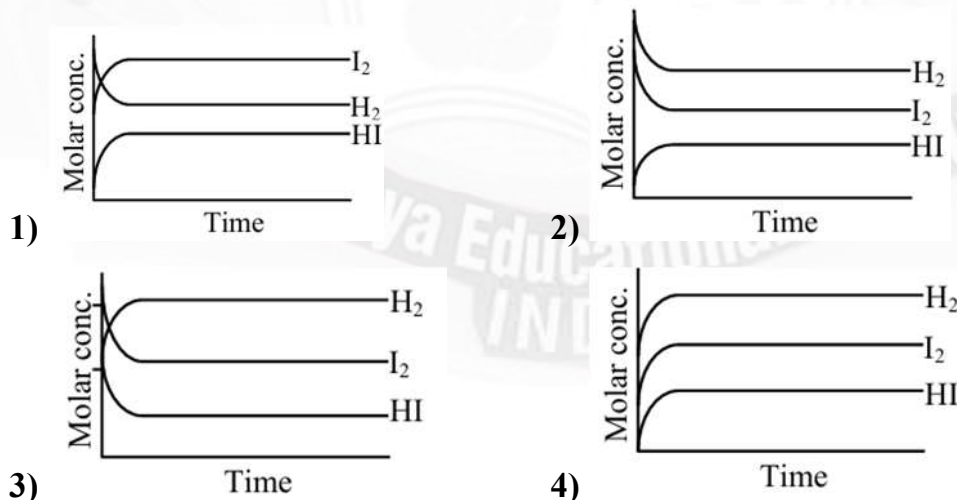


- 1) ZnS 2) CoS 3) MnS 4) NiS

69. For the reaction,



Attainment of equilibrium is predicted correctly by:





70. Match List-I with List-II.

	List-I (Transition metal ion)		List-II (Spin only magnetic moment (B.M.))
(A)	Ti ³⁺	(I)	3.87
(B)	V ²⁺	(II)	0.00
(C)	Ni ²⁺	(III)	1.73
(D)	Sc ³⁺	(IV)	2.84

Choose the correct answer from the options given below :

- 1) (A)-(III), (B)-(I), (C)-(II), (D)-(IV) 2) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
 3) (A)-(IV), (B)-(II), (C)-(III), (D)-(I) 4) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

SECTION-II

(NUMERICAL VALUE TYPE)

This section contains **5 Numerical Value Type Questions**. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. The hydrocarbon (X) with molar mass 80 g mol^{-1} and 90% carbon has _____ degree of unsaturation.
72. In Carius method of estimation of halogen, 0.25 g of an organic compound gave 0.15 g of silver bromide (AgBr). The percentage of Bromine in the organic compound is _____ $\times 10^{-1}\%$ (Nearest integer).
 (Given : Molar mass of Ag is 108 and Br is 80 g mol^{-1})
73. The observed and normal molar masses of compound MX_2 are 65.6 and 164 respectively. The percent degree of ionisation of MX_2 is _____. (Nearest integer)
74. The possible number of stereoisomers for 5-phenylpent-4-en-2-ol is _____.
75. Consider a complex reaction taking place in three steps with rate constants k_1 , k_2 and k_3 respectively. The overall rate constant k is given by the expression $k = \sqrt{\frac{k_1 k_3}{k_2}}$. If the activation energies of the three steps are 60, 30 and 10 kJ mol^{-1} respectively, then the overall energy of activation in kJ mol^{-1} is _____. (Nearest integer)





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